

INSCONE: Unknown-Aware Detection of LLM-Generated Text via Informed Wild Data

Mark Stanley, Samad Syed, Masa Abboud, Saira Khatoon, and Fairoz Khan

University of Wisconsin–Madison

{mgstanley, ssyed9, mabboud, skhatoon, fkhan35}@wisc.edu

Abstract

AI-generated text is increasingly difficult to distinguish from human writing, creating risks in academic integrity, medical misinformation, and social media disinformation. While recent work has reframed MGT detection as an out-of-distribution problem (Zeng et al., 2025) with strong in-distribution and zero-shot results, generalization to LLMs unseen during training remains underexplored. We address this with two contributions. First, we propose **INSCONE** (Informed **SCONE**), which adapts the SCONE wild-data framework (Bai et al., 2025) to the text domain by exploiting curated wild data with known mixing proportions to stabilize the energy geometry around seen and unseen LLM families, achieving strong zero-shot generalization on the RAID benchmark (Dugan et al., 2024). Second, using Open-Set Recognition (OSR) as a base, we propose a 3 Zone Classification, aiming to simultaneously classify known LLMs, unknown LLMs, and human text. We additionally characterize the adversarial robustness of the HTAO detector and extend the RAID benchmark with regenerations from contemporary frontier models, providing a broader empirical picture of detector behavior under realistic deployment conditions.

1 Introduction

AI-generated text is becoming increasingly difficult to distinguish from human writing, with real consequences for academic misuse, medical misinformation, and disinformation on social media. As LLMs become more capable and their outputs more fluent, reliable detection systems become correspondingly more important. The core challenge for any practical detector is *unknown-awareness*: the ability to identify AI-generated text from models never seen during training.

Most existing approaches fall into two categories. Zero-shot methods (DetectGPT, Binoculars, Fast-DetectGPT) exploit statistical patterns in text without labeled training data. Supervised classifiers

(DeTeCtive, RADAR) are trained on labeled human and AI-generated text. Both paradigms assume a fixed model landscape, and Wu et al. (2024) identify cross-model generalization as a central open problem: detectors trained on one model family often fail on newer, unseen ones. Pu et al. (2022) quantify this directly, finding that no single supervised detector generalizes across LLM families and that GPT and LLaMA families are largely isolated from each other in detector space. This is strong empirical motivation for unknown-aware detection: even carefully supervised detectors hit a hard ceiling when faced with unseen models.

We build on the OSR framing of Zeng et al. (2025). We flip the detection question and propose a 3 Zone Classification. Instead of learning what human text looks like, they treat human writing as the outlier. LLM-generated text forms a learnable in-distribution $\mathcal{P}_{\text{in}}$, while human text constitutes semantic OOD $\mathcal{P}_{\text{out}}^{\text{sem}}$, a more heterogeneous class that resists parametric modeling. Under this framing, unseen LLMs represent covariate OOD $\mathcal{P}_{\text{out}}^{\text{cov}}$: same semantic class, shifted distribution. An energy-based detector draws a boundary around AI-generated text and flags anything outside it as human-written.

The natural extension is to incorporate wild data, unlabeled mixtures of covariate and semantic OOD, to improve generalization to unseen LLMs, following Bai et al. (2025). However, we identify a fundamental mismatch: SCONE’s original loss uniformly pushes all wild samples toward high energy, which is benign in vision because covariate shift is pixel-level and the Lipschitz drag from ID samples dominates. In text, “covariate” means a model that has not been seen during training, the embedding distance from ID is large and heterogeneous, and the push term overwhelms the implicit drag, actively degrading zero-shot performance.

Our key insight is that when the wild batch is explicitly curated with known mixing proportions

$(\pi_{\text{id}}, \pi_c, \pi_s)$, the quantile boundary between covariate and semantic OOD is exact and requires no estimation. We call this formulation **INSCONE** (**I**nformed **S**CONe) and implement it as a proximal/distal split in energy space with a dead-zone buffer around the composition boundary. Unlike prior detectors that require retraining for each new model family, INSCONE improves generalization without architectural changes to the base detector.

We further investigate whether the energy surface admits three-zone inference, separating known LLMs, unknown LLMs, and human text simultaneously. We implement four architectures and find that the ID/covariate boundary is the fundamental bottleneck, establishing an encoder-level ceiling on what any energy-based three-zone detector can achieve. We also evaluate HTAO’s adversarial robustness and extend the benchmark with frontier model regenerations not covered by prior work.

Our contributions are: (1) INSCONE, an informed wild-data loss that resolves the covariate/semantic conflation of SCONE for text; (2) a systematic three-zone OSR investigation across four architectures with a characterization of the encoder-level bottleneck; (3) an adversarial analysis separating generation-time from post-hoc attacks on HTAO; and (4) create an extended evaluation set regenerating RAID prompts with contemporary frontier models unavailable for future exploration.

2 Related Work

MGT detection. Zero-shot methods (Mitchell et al., 2023; Hans et al., 2024) and supervised classifiers (Hu et al., 2023; Verma et al., 2024) both assume a fixed model landscape. Pu et al. (2022) train ELECTRA-large classifiers on 13 LLMs and find that detectors trained on medium-sized models generalize within a family but not across families, meaning a universal detector requires either an ensemble or mixed training data. Wu et al. (2024) identify cross-model generalization as the central open problem.

OOD detection for MGT. Zeng et al. (2025) reformulate MGT detection as OSR, combining DeepSVDD, HRN, and energy-based scoring with contrastive pretraining on SimCSE-RoBERTa. The key insight is that LLM-generated text occupies a tighter distributional region than human text, making LLM text the natural in-distribution class. Our work builds directly on their energy-based variant.

Wild data and open-set generalization. Bai et al. (2025) propose SCONE for vision OSR, using unlabeled wild data with an energy margin loss. The theoretical justification relies on a Lipschitz drag argument (Proposition 3.1 of Bai et al. 2025) that assumes small covariate shifts, an assumption violated in the text domain. du Plessis et al. (2015) establish that known class priors enable strictly better label assignment to unlabeled data in PU learning, which motivates our informed quantile split. Chen et al. (2026) propose UnLLM for open-set text classification, reformulating classification as a subset-conditioned generation task that intrinsically models unknowns. Our three-zone OSR investigation is inspired by this line of work, though we find the encoder-level boundary is the binding constraint before such modeling choices matter.

Adversarial robustness. Krishna et al. (2023) show that recursive paraphrasing can evade many MGT detectors. Our evaluation complements this by isolating post-hoc character perturbations as a distinct attack surface not addressed by OSR-based detection.

3 Methods

3.1 Baseline: Energy-Based MGT Detector

Following Zeng et al. (2025), we use a frozen SimCSE-RoBERTa encoder (Gao et al., 2022) (princeton-nlp/unsup-simcse-roberta-base) with a 3-layer MLP classification head over K LLM family classes. The free energy score $E(x) = -\log \sum_i \exp(f_i(x))$ serves as the anomaly score: low energy indicates a confident ID (machine) prediction; high energy flags human or unseen LLM text.

The combined training loss is:

$$\mathcal{L} = \alpha_{\text{con}}\mathcal{L}_{\text{con}} + \alpha_{\text{cls}}\mathcal{L}_{\text{cls}} + \alpha_{\text{eng}}\mathcal{L}_{\text{eng}} \quad (1)$$

where $\mathcal{L}_{\text{con}}$ is a SimCLR-style binary contrastive loss grouping machine vs. human text, $\mathcal{L}_{\text{cls}}$ is cross-entropy over known LLM families on machine text only, and

$$\mathcal{L}_{\text{eng}} = \mathbb{E}[\text{ReLU}(E - m_{\text{in}})^2 \cdot \mathbf{1}_{y=1} + \text{ReLU}(m_{\text{out}} - E)^2 \cdot \mathbf{1}_{y=0}] \quad (2)$$

shapes the energy surface with margins $m_{\text{in}} = -7.0$ (ID anchor) and $m_{\text{out}} = -2.0$ (OOD anchor). At inference, a threshold τ at the 95th percentile of ID validation energy calibrates the FPR95 operating point.

3.2 INSCONE: Informed Wild-Data Loss

Problem setup. To simulate wild data without requiring internet-scraped unlabeled corpora, we use held-out LLMs. Our data is split into three categories: ID models used during standard training, OOD-covariate models used with the wild loss, and unseen LLMs held out entirely for zero-shot testing. We choose these models to balance a temporal scale (ID models are earlier, covariate models are later, zero-shot models are most contemporary) while maintaining at least one model per family for energy classifier coverage. Critically, because the wild batch is explicitly curated, the mixing proportions $(\pi_{\text{id}}, \pi_c, \pi_s)$ are known exactly, and the quantile boundary between covariate and semantic OOD requires no estimation.

Why standard SCONE is insufficient for text. The original SCONE wild loss $\text{ReLU}(-E_{\text{wild}})^2$ pushes all wild samples toward high energy. In vision OSR, this is harmless: covariate shift is pixel-level, and the Lipschitz drag from ID samples pulls nearby covariate samples back down (Bai et al., 2025). In text, covariate shift means a different model family entirely. The embedding distance from ID is large and heterogeneous, so the push term dominates and covariate LLMs are incorrectly driven toward the human text region. Empirically, wild_memorization FPR95 degrades from 0.1942 $\rightarrow$ 0.3848 under standard SCONE relative to the data-matched baseline.

Informed quantile split. Given known mixing proportions, we define proximal and distal thresholds:

$$\tau_{\text{prox}} = \text{Quantile}(E_{\text{wild}}, 1 - \pi_s - \delta/2) \quad (3)$$

$$\tau_{\text{dist}} = \text{Quantile}(E_{\text{wild}}, 1 - \pi_s + \delta/2) \quad (4)$$

where δ is a buffer parameter that creates a dead zone around the composition boundary. The INSCONE loss is:

$$\begin{aligned} \mathcal{L}_{\text{INSCONE}} = & \text{ReLU}(E_{\text{wild}}[E \leq \tau_{\text{prox}}] - m_{\text{in}})^2 \\ & + \text{ReLU}(-E_{\text{wild}}[E \geq \tau_{\text{dist}}])^2 \end{aligned} \quad (5)$$

The proximal term pulls estimated covariate OOD toward the ID energy region; the distal term pushes estimated semantic OOD toward high energy. Samples in $(\tau_{\text{prox}}, \tau_{\text{dist}})$ receive no gradient, preventing spurious updates from compositionally ambiguous samples. At $\delta = 0$, this recovers a clean π_s -split. Because this relies on the wild energy distribution

being ordered as described, we apply a warmup of several epochs before activating the wild loss. The total loss adds $\lambda_{\text{INSCONE}} \mathcal{L}_{\text{INSCONE}}$ to Equation (1).

This approach relies on the assumption that unseen LLM data occupies lower energy than human text under the trained baseline. Figure 1 confirms this empirically: as noted by Zeng et al. (2025), the energy-based detector already generalizes well to unseen models, with zero-shot LLMs tracking the energy distribution of covariate seen models rather than human text. Figure 2 corroborates this at the wild batch level: the quantile split cleanly partitions by class, with $\sim 99.2\%$ of proximal samples being ID or covariate OOD and $\sim 99.6\%$ of distal samples being human text. INSCONE exploits this preexisting geometry to further tighten generalization.

Finally, while our formulation simulates wild data via held-out LLMs, the approach extends naturally to truly unlabeled corpora: a reasonable estimate of $(\pi_{\text{id}}, \pi_c, \pi_s)$ combined with the buffer δ is sufficient to apply the proximal/distal split, making INSCONE applicable beyond the controlled experimental setting.

3.3 OSR Investigation

While INSCONE optimizes binary AUROC, a stricter test of an unknown-aware detector is whether the energy classifier can perform fine-grained three-way classification: KNOWN family, COVARIATE machine text (covering both seen-but-untrained models and entirely unseen models at inference time), and HUMAN. This requires two calibrated thresholds $(t_{\text{mid}}, t_{\text{out}})$ on the energy axis instead of one, and we measure macro Zone Accuracy across three test pillars (each pairing one LLM type with held-out humans).

The vanilla SCONE detector achieves 99% AUROC on every pillar but only 70.6% macro Zone Accuracy. The bottleneck is calibration: when ID and dev_covariate energies overlap, quantile-95 thresholds collapse to $\tau_{\text{mid}} \approx \tau_{\text{out}}$, which leaves an empty band, so covariate samples are never classified as COVARIATE.

First, we replace quantile-95 calibration with a grid search over $t_{\text{mid}} \approx t_{\text{out}}$ on the macro zone-accuracy objective. This requires no retraining and recovers latent zone separability that the trained energies already encode (+18% absolute). Second, we introduce SupervisedZoneDetector (SZD). SCONE’s quantile-based wild zone assignment is unstable at random initialization, since directional

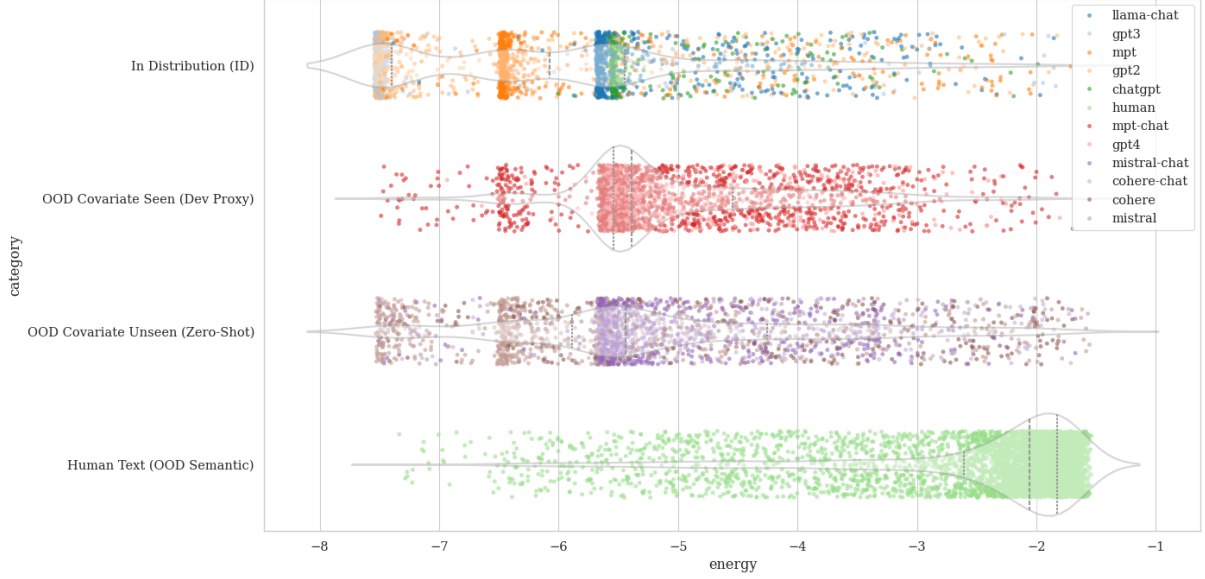


Figure 1: Energy distributions by split under the trained baseline. ID and covariate LLMs occupy similar low-energy regions; human text concentrates near the OOD margin. Zero-shot unseen models track the covariate seen distribution, supporting the temporal generalization hypothesis.

pushes can lock in inverted ordering. In one such run, the wild loss drove machine energies above human energies, giving an AUROC of 0.0016. We replace quantile zones with model-name routed zones (the wild pool is constructed from labeled subsets, so source labels are free at training time), and use directional energy targets: ID samples are pushed to m_{in} , covariate to $(m_{in} + m_{out})/2$, and human to m_{out} . We also raise α_{energy} from 0.01 to 0.1 so the energy margin competes with the K-way classifier (+6% absolute).

Third, we add outlier exposure on covariate logits. SZD memorizes "covariate equals mptchat or gpt4 specifically", so unseen LLMs (mistral, cohere, etc.) cluster near ID rather than in the band. Adding KL(softmax || uniform) on covariate logits forces the head to express genuine uncertainty on covariate samples, which is a representation that generalizes to any unfamiliar machine text (+3% absolute, almost entirely on the unseen pillar).

Fourth, we follow standard OSR protocol by splitting each zero-shot model 50/50 into dev and test partitions. Zero-shot dev energies inform threshold selection; gradients never see them. Zero-shot test is reserved for final reporting.

3.4 Adversarial Robustness Analysis

We evaluate HTAO’s robustness under the threat model implied by its OSR framing: an adversary attempting to generate LLM text that evades detec-

tion. All attacks are black-box with zero knowledge of detector architecture, weights, or threshold. The pretrained HTAO DeepSVDD checkpoint (RAID domain) serves as the target.

Generation-time baseline: 4 persona prompts $\times$ 5 decoding configurations (temperature, top-p, top-k, presence/frequency penalty) applied to Qwen2.5-7B-Instruct, yielding 1,000 samples.

Phase 1 (post-hoc surface transforms): 12 character/token-level perturbations including typo injection, filler insertion, homoglyph substitution, casualization, hedging, and sentence fragmentation, each applied to 1,000 base samples (12,000 total).

Phase 1.5 (stacked, tuned): 8 optimized combinations of Phase 1 winners with tuned perturbation rates (8,000 total).

Phase 2 (LLM rewriting): 5 prompting strategies via Qwen2.5-1.5B-Instruct for semantic-level rewriting (5,000 total).

4 Experimental Setup

Dataset. We use the RAID dataset (Dugan et al., 2024) for all experiments. Our **scone-temporal split** partitions models by approximate release date: ID = {GPT-2, GPT-3, MPT, ChatGPT, LLaMA-Chat}; covariate OOD = {GPT-4, MPT-Chat}; zero-shot test = {Mistral, Mistral-Chat, Cohere, Cohere-Chat}. Human text is semantic OOD throughout.

Due to compute constraints, all experiments are

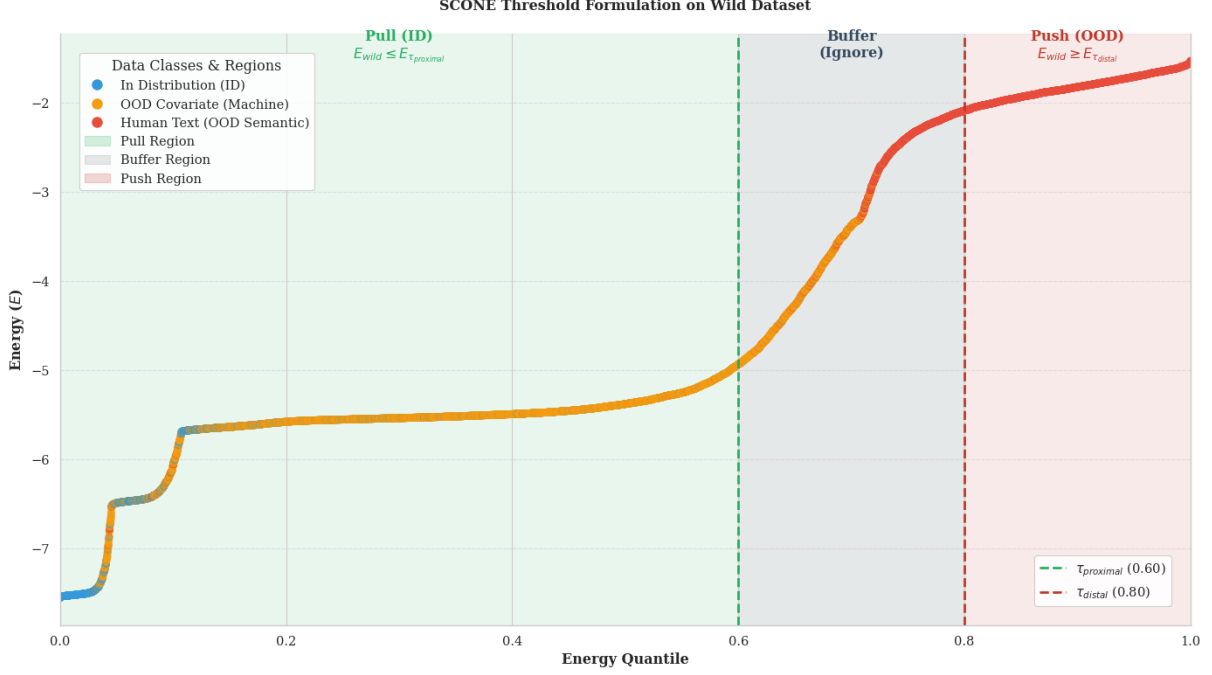


Figure 2: Empirical wild batch energy quantile curve with wild ratios (0.1, 0.6, 0.3) and buffer $\delta=0.1$. The proximal region (green, $E \leq \tau_{\text{prox}}$) anchors covariate machine samples at low energy; the distal region (red, $E \geq \tau_{\text{dist}}$) anchors human text at high energy; the buffer (gray) receives no gradient. Quantitatively, $\sim 99.2\%$ of proximal samples are ID or covariate OOD, and $\sim 99.6\%$ of distal samples are human text, confirming the quantile split partitions by class with minimal contamination.

conducted on a held-out subset of RAID rather than the full benchmark. Results should be interpreted accordingly.

We additionally construct an **extended evaluation set** by regenerating RAID prompts with frontier models not available at HTAO publication: Gemini-2.5-Flash, DeepSeek-V3, and LLaMA-3.3-70B, among others, via LiteLLM ($\approx 2,000$ samples per model, evaluation only).

Compared methods. All variants share the same encoder: SimCSE-RoBERTa-base with intermediate-layer feature fusion (layers 6 and 12), and a 3-layer MLP classification head ($768 \rightarrow 192 \rightarrow 48 \rightarrow K$). We compare four configurations:

- **Baseline:** no wild data, $\lambda_{\text{INSCONE}}=0$. The standard energy detector trained on ID models only.
- **Baseline (fair):** covariate seen models (GPT-4, MPT-Chat) are added to the labeled ID training set. This controls for data access, isolating the effect of the wild-loss formulation from the effect of simply seeing more model families.
- **SCONE:** identical setup to INSCONE but us-

ing the original uniform wild push loss (Bai et al., 2025).

- **INSCONE (ours):** $\lambda_{\text{INSCONE}}=0.005$, $\delta=0.1$, $\pi = (0.1, 0.6, 0.3)$.

Training. AdamW optimizer, cosine LR schedule, bf16-mixed precision, 500-step warmup, SCONE warmup of 4 epochs. Key hyperparameters: $m_{\text{in}}=-7.0$, $m_{\text{out}}=-2.0$, $\alpha_{\text{eng}}=0.001$. Best checkpoint selected by validation AUROC.

Metrics. AUROC and FPR95 (false positive rate at 95% TPR) on the wild_memorization and zero_shot evaluation pillars.

5 Results

5.1 INSCONE vs. Baselines and Standard SCONE

Table 1 reports results on the full-scale RAID scone-temporal split. INSCONE achieves the best zero-shot FPR95 (0.2252), while matching wild-memorization FPR95 (0.1764) and overall AUROC (0.9568) with both baselines.

Two findings are worth highlighting. First, the fair baseline, which adds covariate seen models

Method	AUROC $\uparrow$	Wild $\downarrow$	ZS $\downarrow$
Baseline	0.9577	0.1646	0.2862
Baseline (fair)	0.9571	0.1552	0.3166
SCONE	0.9164	0.3848	0.4270
INSCONE (ours)	0.9568	0.1764	0.2252

Table 1: Main results on RAID (full-scale scone-temporal split). We report overall AUROC, wild FPR95, and zero-shot FPR95. INSCONE achieves the best zero-shot FPR95.

directly to labeled training, does not improve zero-shot FPR95 and slightly degrades it (0.3166 vs. 0.2862). This is consistent with the Pu et al. (2022) finding that supervised classifiers overfit to seen families: adding more data overall causes the classifier to memorize those families rather than generalize to the unseen zero-shot models. INSCONE avoids this by exposing covariate models only through the wild loss, where they are pulled toward the ID energy region without being assigned a family label. Second, standard SCONE is strictly worse than the naive baseline on all three metrics, confirming that the uniform push formulation is not appropriate for the text domain.

Ablation. Table 2 reports a sweep over wild ratios and buffer size on a smaller dataset (3 seeds), used to select the best configuration for the full-scale run. The best zero-shot FPR95 (0.2991) is achieved at $\pi = (0.1, 0.6, 0.3)$, $\delta = 0.1$. At $\delta = 0.0$, boundary samples with ambiguous composition membership receive gradient and introduce noise. At $\delta = 0.2$, the dead zone discards too many informative samples and training becomes unstable. Removing the semantic OOD component entirely ($\pi = (0.0, 1.0, 0.0)$) is the worst configuration, confirming the distal push on human text is a necessary component of the loss. Including or excluding ID samples in the wild batch ($\pi_{id} \in \{0.0, 0.1\}$) makes negligible difference, consistent with their signal being redundant with the labeled training set. However, there was significantly high run variance from different seeds, making it difficult to draw strong conclusions. Overall, all approaches report better results on zero-shot metrics than baseline, implying that better unknown generalization is not dependent on hyperparameter setup, but rather the approach itself.

5.2 OSR Results

Table 3 reports macro Zone Accuracy across successive detector refinements, tracking how each

Ratios (π)	δ	AUROC $\uparrow$	Wild $\downarrow$	ZS $\downarrow$
Baseline (no wild)		0.9175	0.6094	0.5860
Baseline (fair)		0.9224	0.1942	0.4228
(0.1, 0.6, 0.3)	0.0	0.9345	0.2265	0.3166
(0.1, 0.6, 0.3)	0.1	0.9400	0.1827	0.2991
(0.1, 0.6, 0.3)	0.2	0.9406	0.2506	0.3802
(0.0, 0.7, 0.3)	0.0	0.9316	0.2440	0.3188
(0.0, 1.0, 0.0)	0.0	0.9116	0.3337	0.4142

Table 2: Hyperparameter ablation on small-scale RAID runs. We report the average **Overall AUROC**, **Wild FPR@95**, and **Zero-Shot FPR@95** across 3 seeded runs on a very small subset of RAID. The low amount of epochs and data explains general worse performance than reported baseline results. $\delta=0.1$ with $\pi = (0.1, 0.6, 0.3)$ is the best configuration and is used for the full-scale run in Table 1. Removing semantic OOD ($\pi_s=0$) collapses performance; $\delta=0.2$ destabilizes training.

Method	Known	Covariate
Baseline (q-95 calibration)	0.9433	0.5000
SCONE (q-95 calibration)	0.9620	0.6473
+ grid-search calibration	0.6567	0.9827
+ SupervisedZoneDetector (SZD)	0.7593	0.9927
+ Outlier Exposure (SZD-OE)	0.8360	0.9953

Table 3: OSR results across successive detector refinements. Grid-search calibration recovers latent zone separability already encoded in the trained energies (+18% absolute, no retraining). SupervisedZoneDetector (SZD) replaces unstable quantile-based wild assignment with model-name-routed zones and directional energy targets (+6% absolute). Adding outlier exposure on covariate logits encourages uncertainty as a transferable representation for unseen LLMs (+3% absolute). Final calibration follows standard OSR protocol using held-out dev-unseen energies. The stack of four largely orthogonal interventions improves macro Zone Accuracy from 64.8% to 93.2%.

intervention addresses a distinct failure mode in three-zone classification.

5.3 Adversarial Robustness

Table 4 summarizes adversarial results. Generation-time attacks achieve 0% evasion, validating HTAO’s core thesis that the OOD decision surface is robust to how text is generated. Phase 1 post-hoc surface transforms achieve up to 15.4% individual evasion (fillers); optimized stacking (Phase 1.5) pushes the best combination (typos + fillers, blunt persona) to 96% evasion on the base set.

The 96% fool rate from stacked typo and filler injection warrants careful interpretation. These perturbations corrupt subword tokenization in ways

Phase	Fool Rate	Best Combo
Generation-time	0.0%	N/A
Phase 1	10.1%	70%
Phase 1.5	16.1%	96%
Phase 2	5.8%	26%

Table 4: Adversarial results against HTAO DeepSVDD (RAID domain), all black-box. Generation-time attacks achieve 0% evasion; stacked post-hoc surface transforms reach 96% on the best single combination.

that shift RoBERTa embeddings independently of semantic content. This is not a failure of the OOD decision surface: a human reader can trivially identify such texts as machine-generated and then corrupted. Rather, it exposes a tokenizer-level brittleness orthogonal to the semantic OOD framing. LLM rewriting (Phase 2) achieves substantially lower evasion (5.8% average), with the exception of `fragment_stream` (12.7%), suggesting that semantic-level rewrites are harder to evade but also harder to construct effectively.

6 Conclusion

We introduced INSCONE, an informed wild-data loss that resolves a fundamental mismatch between SCONE’s vision-domain assumptions and the structure of LLM embedding space. By exploiting known wild batch composition, INSCONE applies targeted proximal pulls and distal pushes rather than uniform energy pushing, achieving the best zero-shot FPR95 (0.2252) across all compared methods while maintaining strong overall AUROC. The fair baseline comparison further shows that simply adding covariate models to labeled training does not improve zero-shot generalization and can degrade it, confirming that the wild-loss formulation is the source of INSCONE’s gains.

We further evaluate the energy detector under three-zone classification (KNOWN family, COVARIATE machine text, HUMAN), measured as macro Zone Accuracy across three test pillars. Vanilla SCONE achieves 99% AUROC on every pillar but only 70.6% macro Zone Accuracy because the band between the two calibrated thresholds collapses when ID and dev covariate energies overlap. Stacking grid-search calibration, a supervised model-name-routed wild loss, outlier exposure on covariate logits, and held-out dev unseen calibration lifts macro Zone Accuracy to 93.2% with no zone below 84%, suggesting the four interventions are largely orthogonal and each addresses

a distinct failure mode.

Our adversarial analysis separates generation-time attacks (fully repelled by HTAO’s OOD surface) from post-hoc surface perturbations (highly effective but semantically detectable), suggesting certified tokenizer robustness is a necessary complement to OOD-based detection. Future directions include stronger backbones with explicit family-level structure, adversarial evaluation of INSCONE, and public release of the extended frontier model evaluation set.

Limitations

INSCONE assumes that wild batch composition (π_{id}, π_c, π_s) is known or approximately known at training time, which requires deliberate data curation, or reliable estimations of wild data.

Adversarial results are evaluated exclusively against the HTAO checkpoint and should not be extrapolated to INSCONE without additional experimentation.

Future Work

Future work should investigate whether INSCONE generalizes to other OOD domains beyond MGT detection, such as out-of-distribution detection in vision or tabular settings where wild data composition can similarly be curated. On the MGT side, a natural extension is adversarial evaluation of INSCONE directly, rather than the HTAO checkpoint evaluated here, to determine whether the tighter energy geometry confers robustness to post-hoc surface perturbations. Finally, the extended frontier model evaluation set could be released publicly to support reproducible benchmarking of detectors against contemporary LLMs.

Contributions

Masa Abboud and Saira Khatoon: constructed the extended evaluation set by regenerating RAID prompts with contemporary frontier models (Gemini-2.5-Flash, DeepSeek-V3, LLaMA-3.3-70B) via LiteLLM. This involved prompt engineering, API integration across multiple providers, data cleaning, and uploading the resulting dataset to HuggingFace for public release.

Mark Stanley: Led the project’s core technical contribution. Formulated the INSCONE loss, including the proximal/distal quantile split and buffer mechanism, and implemented the full training pipeline in PyTorch Lightning. Reproduced the

HTAO energy-based baseline, designed the scone-temporal RAID split, and conducted the full-scale and ablation experiments. Also responsible for the energy distribution analysis and visualizations (Figures 1 and 2).

Fairoz Khan: designed and executed the adversarial robustness evaluation pipeline. Implemented all attack phases, including post-hoc surface transforms (typo injection, filler insertion, homoglyph substitution, casualization), stacked combinations, and LLM-based rewriting strategies, and analyzed fool rates across generation-time and post-hoc attack surfaces.

Samad Syed: led the three-zone OSR investigation. Designed and implemented four detector architectures (EnergyDetector, MahalEnergyDetector, SupervisedZoneDetector, and K+1 Detector), developed the grid-search calibration procedure that recovers latent zone separability without re-training, introduced outlier exposure on covariate logits to improve generalization to unseen LLM families, and established the held-out dev-unseen calibration protocol following standard OSR practice.

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A Hyperparameter Details

Hyperparameter	Value
Encoder	princeton-nlp/unsup-simcse-roberta-base
Head dims	$768 \rightarrow 192 \rightarrow 48 \rightarrow K$
m_{in}	-7.0
m_{out}	-2.0
α_{con}	1.0
α_{cls}	1.0
α_{eng}	0.001
$\lambda_{INSCONE}$	0.005
Buffer δ	0.1
Wild ratios π	(0.1, 0.6, 0.3)
SCONE warmup	4 epochs
LR schedule	cosine, warmup 500 steps
Optimizer	AdamW
Precision	bf16-mixed

Table 5: Training hyperparameters for the full-scale INSCONE run.

B RAID Temporal Split Details

Split	Models
ID (train)	GPT-2, GPT-3, MPT, ChatGPT, LLaMA-Chat
Covariate OOD	GPT-4, MPT-Chat
Zero-shot (test)	Mistral, Mistral-Chat, Cohere, Cohere-Chat
Frontier (eval)	Gemini-2.5-Flash, DeepSeek-V3, LLaMA-3.3-70B
Semantic OOD	Human text (all splits)

Table 6: RAID scone-temporal split. All experiments use 10,000 labeled training samples and 10,000 wild samples.

C Per-Seed Ablation Results

Full per-seed breakdown of the ablation reported in Table 2, included to document run variance across configurations.

Ratios (π)	δ	Seed 42			Seed 43			Seed 44		
		ZS	Wild	AUC	ZS	Wild	AUC	ZS	Wild	AUC
Baseline (no wild)		0.416	0.698	0.916	0.798	0.874	0.903	0.544	0.256	0.934
Baseline (fair)		0.577	0.205	0.923	0.349	0.182	0.927	0.343	0.195	0.918
(0.1, 0.6, 0.3)	0.0	0.275	0.166	0.951	0.346	0.330	0.944	0.328	0.183	0.909
(0.1, 0.6, 0.3)	0.1	0.243	0.139	0.944	0.340	0.233	0.943	0.314	0.176	0.934
(0.1, 0.6, 0.3)	0.2	0.257	0.189	0.939	0.277	0.174	0.951	0.606	0.389	0.932
(0.0, 0.7, 0.3)	0.0	0.243	0.190	0.944	0.386	0.359	0.921	0.327	0.183	0.930
(0.0, 1.0, 0.0)	0.0	0.441	0.480	0.922	0.461	0.321	0.896	0.341	0.200	0.917

Table 7: Per-seed ablation results. ZS = zero-shot FPR95, Wild = wild_memorization FPR95, AUC = AU-ROC. The averaged results are reported in Table 2.